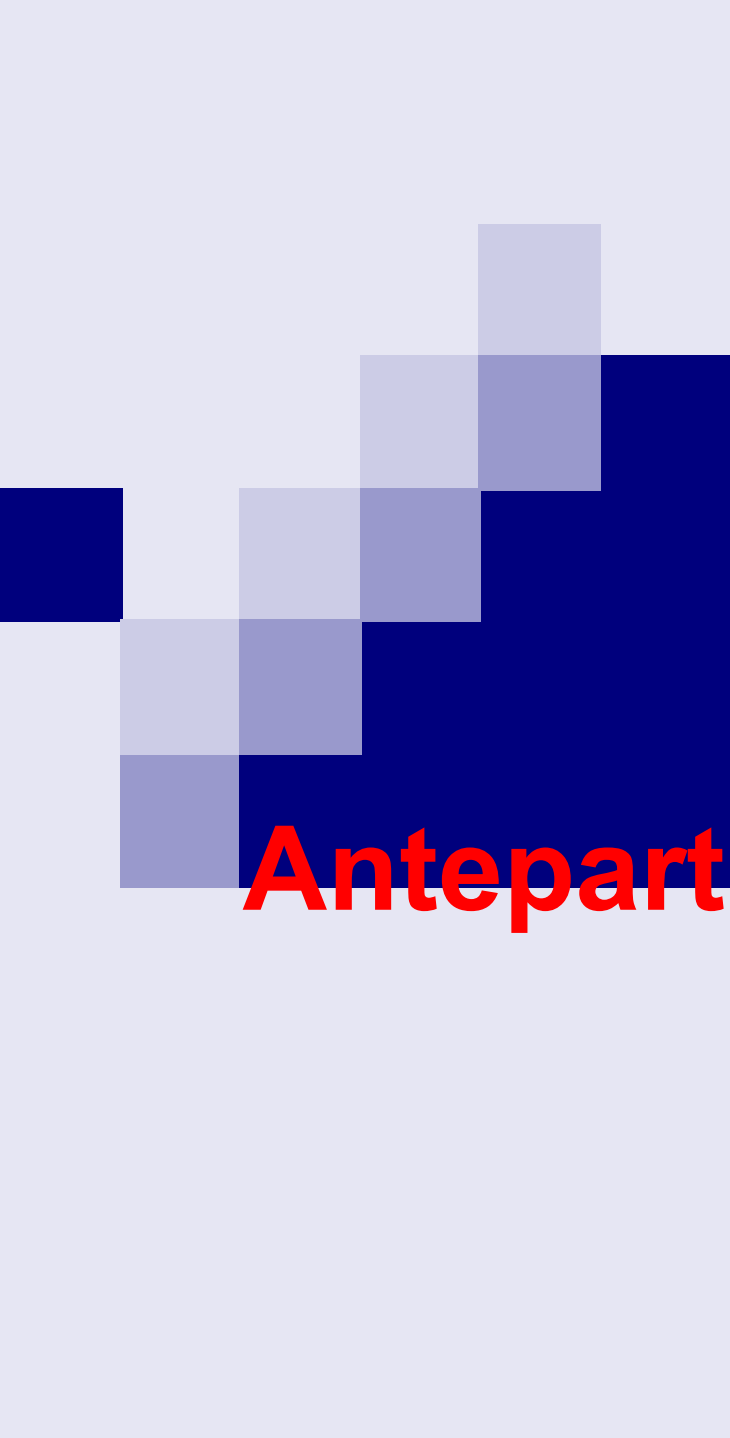




Antepartum Haemorrhage

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Definition

this is defined as vaginal bleeding after **20** completed weeks up to before delivery

Complicates **2-5%** of pregnancy

- and most cases involve relatively small amounts of blood loss. However, significant blood loss poses a risk of mortality and morbidity to both the mother and baby.

Causes

Placental causes:

1. Placenta previa (30%).
2. Abruptio placenta (34%).

Fetal cause

Vasa previa (1%).

Maternal causes (5%):

1. Cervicitis.
2. Cervical erosion.
3. Cervical carcinoma.
4. Vaginal infection.
5. Vaginal trauma.

In 30% of cases no cause can be found.

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- Placental causes are the most worrying, as potentially the mother's and/or fetus's life is in danger and often the bleeding may be more severe than with other causes such as cervical ectropion.
 - However, any antepartum haemorrhage must always be taken seriously, and any woman presenting with a history of fresh vaginal bleeding must be investigated promptly and properly.



Approach to APH HISTORY

- How much bleeding is there?
- What were the triggering factors (e.g. post-coital bleed)?
- Is it associated with pain or contractions?
- Is the baby moving?
- When was the last cervical smear (date and normal/abnormal)?



Approach to APH Examination

- Take the pulse and blood pressure.
- Is the uterus soft or tender and firm?
- Undertake fetal heart auscultation/CTG.
- Undertake speculum vaginal examination, with particular importance placed on visualizing the cervix (having established that placenta is not a praevia, preferably using a portable ultrasound machine).



Approach to APH Investigations

- Depending on the degree of bleeding, full blood count, clotting and, if suspected praevia/ abruption, cross-match 6 units of blood.
- Ultrasound (fetal size, presentation, amniotic fluid, placental position and morphology).

MANAGEMENT

- If there is minimal bleeding and the cause is clearly local vaginal bleeding, symptomatic management may be given (for example antibiotic for cervicitis) as long as there is reasonable certainty that cervical carcinoma is excluded by smear history and direct visualization of the cervix.

Placenta previa

- Means implantation of the placenta in the lower uterine segment.

It complicate **0.5%** of pregnancies(1 in 200 pregnancies).

Bleeding from placenta previa account for about 30% of all cases of APH.

Placenta previa

Predisposing factors:

1. Multiparity.
2. Previous C/S scar.
3. Increasing maternal age.
4. Multiple gestation.
5. Prior placenta previa: patient has a 4% - 8% risk of having p.p. in subsequent pregnancy.
6. Congenital anomaly of the uterus e.g. septate uterus.

What is the lower uterine segment?

- **The lower segment can be defined as that part of the uterine wall which:**
- Does not contract in labour but is stretched in response to contractions.
- Used to be the isthmus before pregnancy.
- Underlies the loose fold of peritoneum that reflects from the bladder.
- Is covered by a full bladder anteriorly.
- Is within **8 cm** of the internal cervical os at term.

Grading of placenta pravia

- Grade .1 (lateral placenta): The placenta implanted in the lower uterine segment but does not reach the internal os.
- Grade .2.(marginal placenta):
 - The edge of the placenta reaches the internal os but not covering it.
- Grade.3.(partial placenta previa):
 - The placenta partially covering the internal os.
- Grade.4.(complete placenta previa): The placenta cover the internal os completely.

Grade 1&2 called minor P.P. grade 3&4 major P.P.





Grade I



Grade II

Minor



Grade III



Grade IV

Major



Clinical presentation

1. Painless, recurrent unprovoked vaginal bleeding:

- Placenta praevia characteristically present with unprovoked painless bleeding.
- Bleeding occurs as a result of disruption of the placental attachment secondary to the development and thinning of the lower uterine segment.

Clinical presentation

- In general those with complete previa bleeds earlier and more heavily than those with a partial or minor placenta previa.
- -The bleeding is usually recurrent.
- -The mean gestational age at the onset of the first bleeding is **30 weeks** with one third presenting before 30 weeks.

Clinical presentation

2. malpresentation:

On abdominal examination it is common to find malpresentation in association with placenta previa **(which are either breech or transverse lie in about 35%)**.

- Slight but inconsistent deviation of the presenting part from the midline and difficulty with palpating the presenting part.
- The abdomen is usually **soft** and the fetal heart is normal.

Diagnosis

- Placenta previa is almost exclusively diagnosed nowadays by ultrasound.
- An ultrasound scan will show the position of the placenta clearly within the uterus.
- If the placenta lies in the anterior part of the uterus and reaches into the area covered by the bladder, it is known as a low-lying placenta **(before 24 weeks)** and placenta praevia after **24 weeks**
- About **5%** of patients have some degree of placenta previa on ultrasonic examination before **20 week's** gestation.
- With the development of the lower uterine segment, a relative upward placental migration occurs, with **90%** of these resolves by the 3rd trimester.

Diagnosis

Clinical diagnosis of placenta previa.

- Clinical diagnosis of placenta previa can be made by palpating the placenta through the cervical os .
- ❖ **However this examination can precipitate profuse vaginal bleeding and it is** only indicated when U/S is not available and the patient in labour with non-life threatening vaginal bleeding.

Per vaginal examination is contraindicated.

Such examination should always be done in the theatre with every thing is prepared for caesarian section with a complete operating team ready to operate should vaginal examination precipitate substantial bleeding.

ASYMPTOMATIC LOW-LYING PLACENTA:

- All women with a low-lying placenta diagnosed in early pregnancy should be rescanned at **34 weeks' gestation.**
- **There is no need to restrict work activities or sexual intercourse in women with a low-lying placenta on ultrasound unless they bleed.**
- If the placenta praevia is still present at 34 weeks' gestation and is Grade I or II, the woman should be rescanned on a fortnightly basis but need not be admitted unless they bleed.



Management of patient with bleeding

- **As with any 3rd trimester bleeding:**
- The patient condition should initially be stabilized.
- Fetal monitoring instituted.
- Blood studies ordered.
- Blood products made available.

Management of patient with bleeding

Admit to hospital. Call for help

Insert two large-bore (14 gauge) intravenous cannulae

Urgent venepuncture (20 mls) for:

- Cross match for minimum of 4 units
- Full blood count - Coagulation screen including fibrinogen- Renal and liver function tests
- Until blood is available, infuse up to 3.5 litres of warmed fluids. Initially 2 litres of warmed isotonic **crystalloid** followed by a further 1.5 litres of warmed **colloid** if blood is still not available.
- Insert indwelling urinary catheter with hourly urometer bag.
- Monitor fluid balance.

Managment

- Once the diagnosis of placenta previa is established, management decision will depend on:
 - 1. Gestational age.
 - 2. Severity of bleeding

Preterm

- The aim is to **obtain fetal maturation without compromising the mother's health.**

1-if the bleeding is excessive:

Delivery must be accomplished by caesarian section regardless the gestational age.

2. When the bleeding is not profuse:

- **The patient is managed expectantly in hospital on bed rest and blood transfusion if the woman is anaemic.**
- After several days without bleeding, she may be ambulate and even discharged if she lives nearby.
- Instruct the patient to return at the first sign of further bleeding.
- her haemoglobin should be not less than **11gm.**
- Blood should always be available for the patient.

Term

- **At 36–37 weeks' presentation, a final ultrasound should be performed and acted upon:**
- **(a) Grades II , III and IV placenta praevia** should have a Caesarean section between 37 and 38 weeks' gestation by an experienced obstetrician particularly if the placenta is on the anterior wall of the uterus.
- **(b) If the presenting part is below the lower edge of the placenta in Grade I ,** then it is safe to wait until labour and these women can be expected to deliver vaginally.

Management

- There is increased risk of **feto-maternal transfusion and immunization.**
- So **anti-D immunoglobulin** should be given.
- **A kleihauer-Betke** test should be done on **maternal blood** to determine the extent of the **feto-maternal transfusion** so that an appropriate larger dose of anti D may be given.

Complications maternal

- There is increased maternal mortality and morbidity
 - Higher CS rate
 - Major blood transfusion
 - Longer hospital stay .
 - PPH
 - Renal failure
 - Death

Maternal complications

- **major cause of death in women with placenta praevia now is**

1- postpartum haemorrhage (PPH).

PPH is common because the lower segment does not contract and retract as in the upper segment, and therefore maternal vessels of the placental bed may continue to bleed after delivery. this may lead to an emergency hysterectomy if the bleeding can not be stopped.

2.Associated placenta accreta.

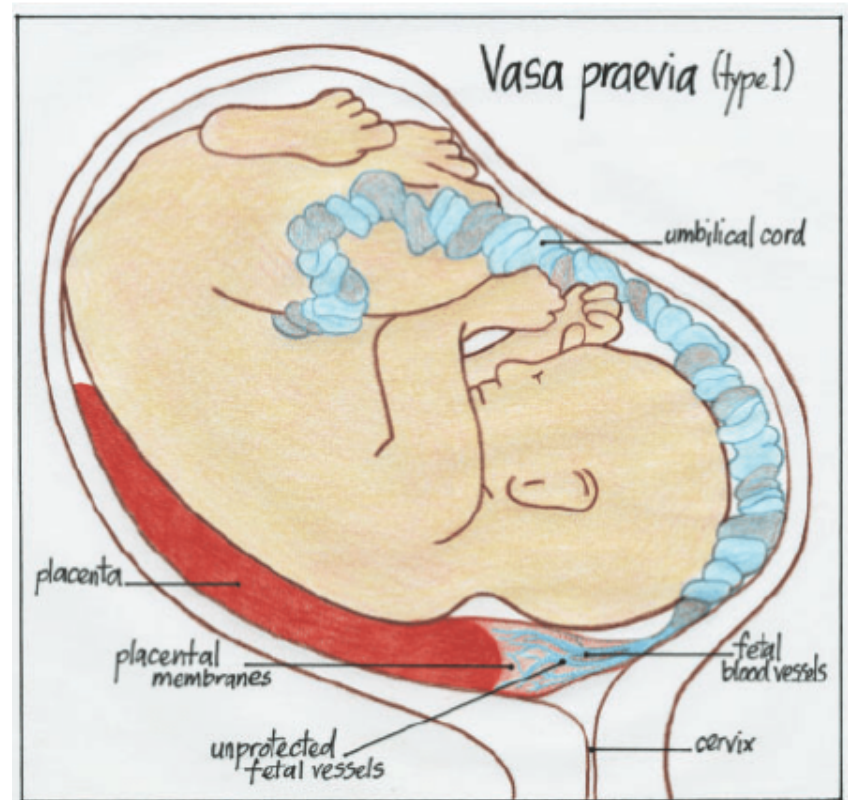
Fetal complications

The perinatal mortality of patients with placenta previa is higher than the general population and this is related to:

1. Prematurity (which is the main cause).
2. Higher incidence of IUGR (about **20%** of pregnancies with placenta previa develops IUGR).
3. Malpresentation (in **30%** of cases).
- 4- The presence of vasa previa which carry a perinatal mortality of **75%**.

vasa previa

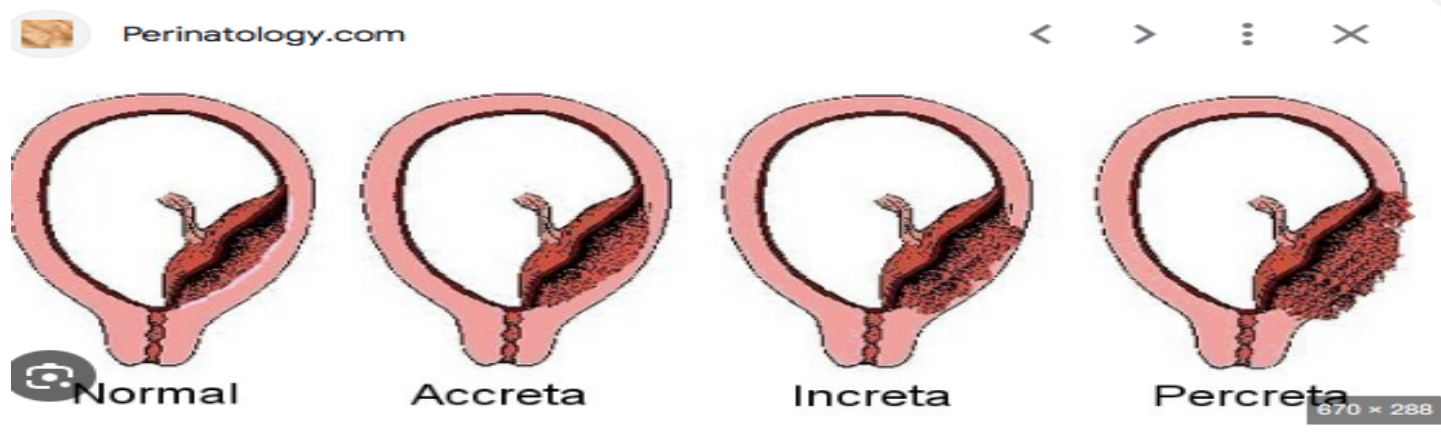
- This is a rare condition
Velamentous insertion of the umbilical cord in the membranes.
- At the time of rupture of membranes (whether spontaneous or artificial) the umbilical vessels will rupture causing massive bleeding which is of fetal origin.
- .It is suspected when fetal heart shows sever bradycardia after rupture of membranes.
- Treatment is by immediate C/S.



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- *Prevention:* avoidance of non-clinically indicated caesarean section

placenta accreta.

- Abnormal attachment of the placenta through the myometrium as a result of defective decidua formation.
- 1-It may be superficial ----- **placenta accreta**
- 2.The placental villi may invade partially through the myometrium-----
- **placenta increta**.
- 3.The villi may invade the serosa----- **placenta percreta**.



placenta accreta.

Risk factors for placenta accreta.

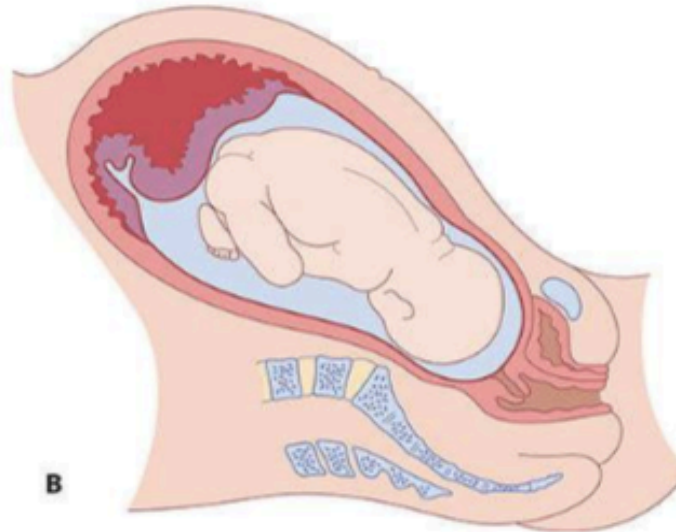
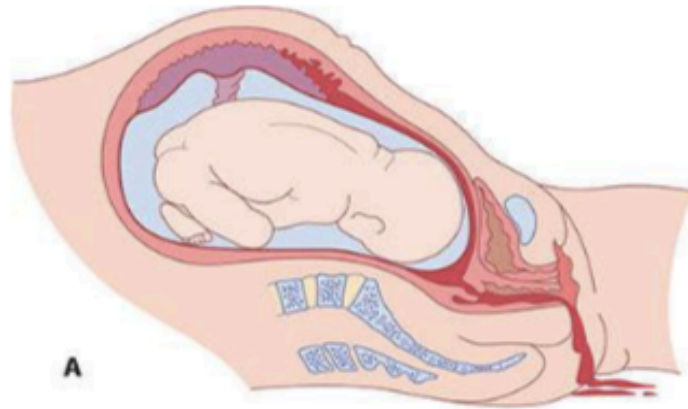
1. Previous uterine surgery.
2. Placenta previa.
3. Congenital anomalies of the uterus.

There will be difficulty in delivering the placenta with massive bleeding.

- **Two third** of patients will require hysterectomy to save their life.

Abruptio placenta

- premature separation of a normally sited placenta.
 - ❖ It complicates **0.5-2%** of pregnancies.
- ❖ It could be of two types:
- **1.Revealed or external bleeding** when there is obvious external vaginal bleeding
 - -This is seen in **two third** of the cases.
- **2.Concealed bleeding** when the bleeding remain in the uterus with no external bleeding.
 - -This is seen in **one third** of the cases.



Causes

- 1.**Idiopathic**:In the majority of cases the cause is unknown.
- 2.There is an association with defective trophoplastic invasion, as with **pre-eclampsia** and **intrauterine growth restriction**.
- 3.**Direct trauma** e.g. road traffic accidents and external cephalic version.
- 4.**High parity**.
- 5.**Sudden decompression of the uterus** e.g after delivery of 1st twin or release of polyhydramnios.
- 6.**Hypertension**.
- 7.**Smoking**.

Clinical presentations

The classical presentation is that of **painful** Vaginal bleeding. associated with a **tense rigid abdomen**.

The **absence of a tense abdomen** does **not rule out** a placental abruption.

- Also, **as the bleeding may be concealed ,it's absence does not exclude the diagnosis.**
- -Abruptio placenta usually occurs near term and frequently during labour.

Clinical presentations

- **Abdominal examination:**
- **tenderness over the abdomen** **woody hard**
- *May be larger than date.
- **The fetus is difficult to palpate.**
- **Fetal distress FH bradycardia**
- **Dead fetus**
- **signs of hypovolumic shock: tachycardia , hypotension , sweating, absence or reduced fetal movements and tense painful abdomen**



Clinical presentations

- **Vaginal examination:**
- ❖ There is vaginal bleeding.
- ❖ Cervical dilatation if the patient in labour.

Diagnosis

- ❖ This is usually made on clinical grounds.
- ❖ U/S can be helpful in some cases, demonstrating retro placental clot and excluding placenta previa.
- but the absence of any ultrasound changes does not rule it out and patients should be managed on the basis of their clinical findings.

Management

Admit to hospital. Call for help

Insert two large-bore (14 gauge) intravenous cannulae

Urgent venepuncture (20 mls) for:

- Cross match for minimum of 4 units
- Full blood count - Coagulation screen including fibrinogen-
Renal and liver function tests
- Until blood is available, infuse up to 3.5 litres of warmed fluids. Initially 2 litres of warmed isotonic **crystalloid** followed by a further 1.5 litres of warmed **colloid** if blood is still not available.
- Insert indwelling urinary catheter with hourly urometer bag.
- Monitor fluid balance.

Management

- **Hypovolumic shock:**
- Usually there is underestimate of the blood lose, due to some haemorrhage being concealed behind the placenta and within the uterine wall.
- Also some patients are hypertensive prior to the abruption, masking the hypotensive effect of blood lose.

Management

- **The management depends on:**
 1. The severity of bleeding.
 2. The gestational age.
 3. The fetal and maternal condition.
- **For all but the most minor degrees of placental abruption, the amount of blood lost from the maternal circulation is likely to be significant.**

- The plasma volume should be monitored by CVP measurement.
- Strict fluid balance.
-  Hourly measurement of urine output (after urethral catheterization).

- 
- **1. When abruption results in fetal death, the management include:**
 - Maternal resuscitation.
 - Induction of labour to achieve vaginal delivery.
 - Caesarian section performed when vaginal bleeding is uncontrollable in order to save the mother's life.

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- **The women can be delivered vaginally if:**
 - 1.The fetal heart is normal (no signs of distress).
 - 2.The women is already in labour or rapid delivery is anticipated if induction tried.

Caesarian section should be done if :

- 1.There is maternal and /or fetal distress.
- 2.The women is not in labour.

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- When there is minor degrees of abruption and the maternal and fetal condition is stable especially when the fetus still premature.
 - This require close monitoring of fetal well being to detect any signs of deterioration in fetal condition.

Maternal Complication

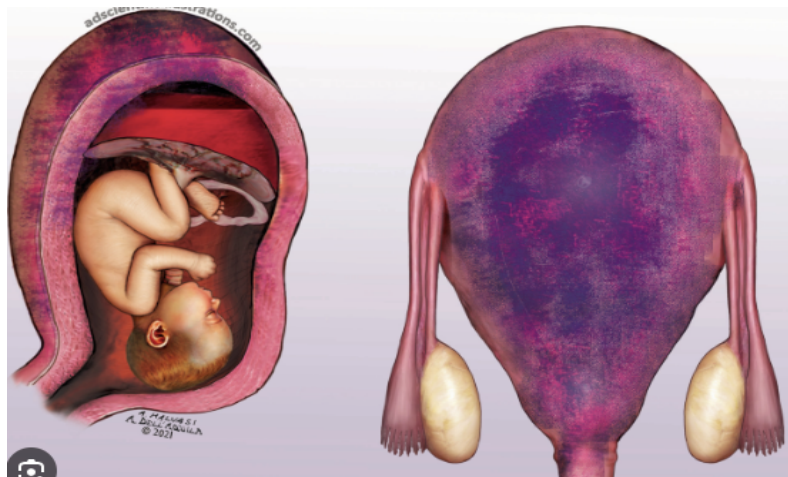
- DIC is always a secondary phenomenon following a trigger to generalized activation of the coagulation system.
- This activation will result in consumption of fibrin, clotting factors and platelets, resulting in continued bleeding and further depletion of these factors.
- ✨ This is due to poor renal perfusion secondary to hypovolaemia, hypotension and DIC.
- ✨ Initially there is oliguria. If reduced renal perfusion persists, the patient will develop acute tubular necrosis.

Maternal Complication

PPH

Couvellaire uterus:

- Extravasations of blood into the myometrium causing purplish discoloration of the uterus. Such uterus does not contract well after delivery leading to PPH.



Maternal Complication

- Death Placental abruption is a significant cause of maternal death as a consequence of these complications.
- Recurrence:
 - ✦ After single attack of abruption the recurrence risk is approximately **10%**
 - ✦ And **25%** after two episodes.

Fetal complications

- **Perinatal mortality:**
- Placental abruption is a significant cause of fetal and neonatal loss.
- **Perinatal mortality rate is affected by:**
 - 1.The size of abruption.
 - 2.Interval to delivery.
 - 3.Gestational age at which abruption and delivery happened.
 - 4.Other factors such as growth retardation due to poor placentation.



General considerations

- **APH is unpredictable and at any time the patient's condition may deteriorate rapidly.**
- **The management must therefore aim to treat or prevent such deterioration.**

Per-vaginal examination

- **This is usually contra-indicated unless placenta previa has been excluded and should be confined to vulval inspection only to assess quickly the amount of blood loss and to determine whether bleeding has stopped or is continuing.**



Thank you